



US012414582B2

(12) **United States Patent**
Lee et al.

(10) **Patent No.:** **US 12,414,582 B2**
(45) **Date of Patent:** **Sep. 16, 2025**

(54) **JUICER**

(56) **References Cited**

(71) Applicant: **ANGEL CO., LTD.**, Busan-si (KR)

U.S. PATENT DOCUMENTS

(72) Inventors: **Mun Hyon Lee**, Nonsan-si (KR);
Kwang Hui Lee, Busan-si (KR)

5,396,836 A * 3/1995 Kim B30B 11/241
241/101.2
8,863,656 B2 * 10/2014 Trovinger A23N 1/02
100/145
10,182,589 B2 * 1/2019 Lee A23N 1/00
2007/0133342 A1 * 6/2007 Gili A47J 43/06
366/129
2007/0269564 A1 * 11/2007 Bertocchi A23L 2/04
426/489
2008/0066630 A1 * 3/2008 Siu A47J 43/082
99/510
2009/0057235 A1 * 3/2009 Sugaya C02F 11/147
210/709

(73) Assignee: **ANGEL CO., LTD.**, Busan-si (KR)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 339 days.

(21) Appl. No.: **16/899,601**

(Continued)

(22) Filed: **Jun. 12, 2020**

FOREIGN PATENT DOCUMENTS

(65) **Prior Publication Data**

US 2021/0274827 A1 Sep. 9, 2021

KR 10-0334309 B1 5/2002
WO WO2013069828 * 5/2013

(30) **Foreign Application Priority Data**

Mar. 5, 2020 (KR) 10-2020-0027696

OTHER PUBLICATIONS

CN 204133101 (Year: 2023).*

(Continued)

(51) **Int. Cl.**

A23N 1/02 (2006.01)
A47J 19/02 (2006.01)
B02C 18/24 (2006.01)
B02C 23/10 (2006.01)

Primary Examiner — Elizabeth M Kerr

Assistant Examiner — Keith Brian Assante

(74) *Attorney, Agent, or Firm* — Insight Law Group,
PLLC; Seung Lee

(52) **U.S. Cl.**

CPC **A23N 1/02** (2013.01); **A47J 19/025**
(2013.01); **B02C 18/24** (2013.01); **B02C 23/10**
(2013.01)

(57)

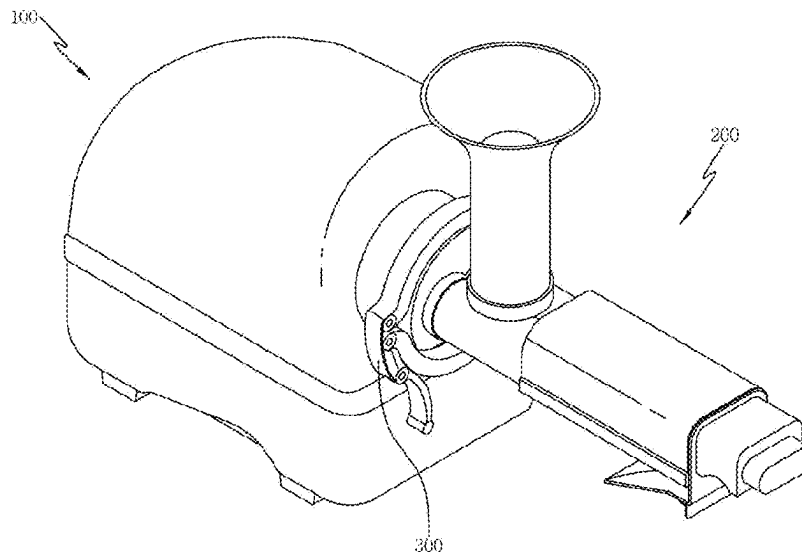
ABSTRACT

The present disclosure relates to a juicer with improved safety, and particularly, includes a motor housing, a drive part including a motor which is provided inside the motor housing, and a roller part which is detachably coupled to extend forward from the motor housing, and is provided therein to be rotatably driven by the drive force transferred from the drive part, in which the drive part and the roller part are attached and detached by the normal and reverse rotation of the motor, thereby greatly improving safety of the juicer.

(58) **Field of Classification Search**

CPC . A23N 1/02; A47J 19/025; A47J 19/06; A47J
19/027; B02C 18/24; B02C 23/10
USPC 99/510
See application file for complete search history.

4 Claims, 9 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2013/0160658	A1 *	6/2013	Torrise		A23N 1/02
					99/511
2014/0245906	A1 *	9/2014	Lee		A23N 1/00
					99/510
2016/0243186	A1 *	8/2016	Mandau		A23L 33/105
2017/0367398	A1 *	12/2017	Cui		A23N 1/02
2021/0039893	A1 *	2/2021	Pettersson		B30B 9/3096
2021/0337767	A1 *	11/2021	Trottier		G09B 7/00

OTHER PUBLICATIONS

EP2777449 (Year: 2023).*

KR1020120127705 (Year: 2023).*

CN203436142 (Year: 2023).*

KR19950026408 (Year: 2023).*

* cited by examiner

FIG. 1

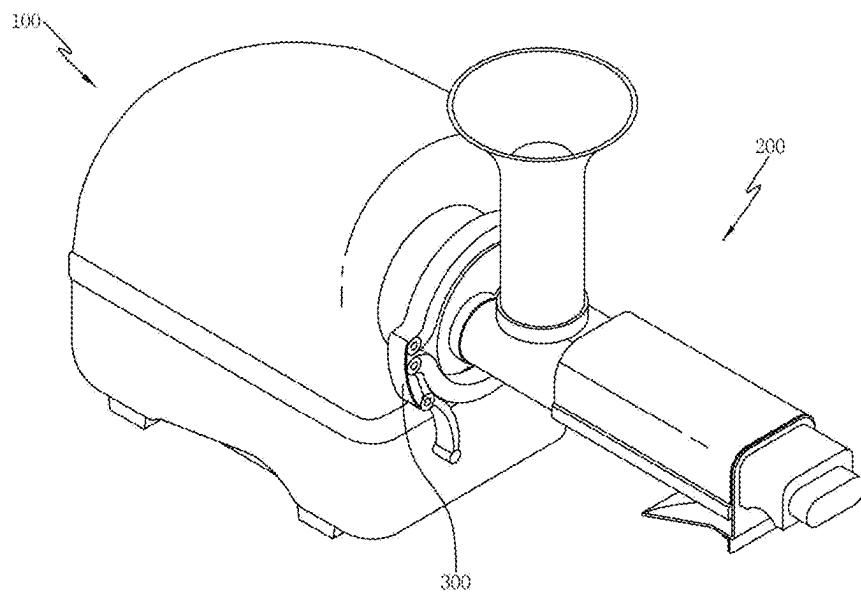


FIG. 2

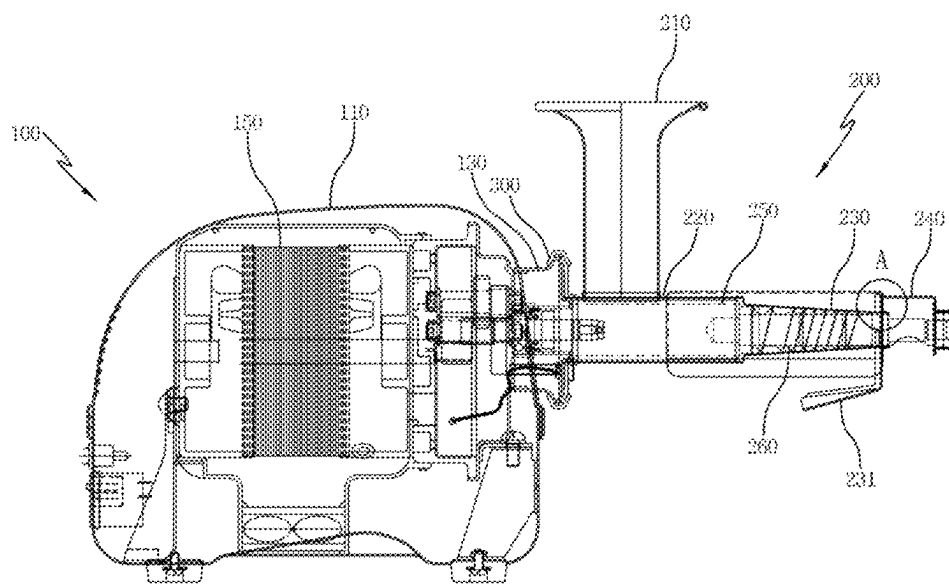


FIG. 3

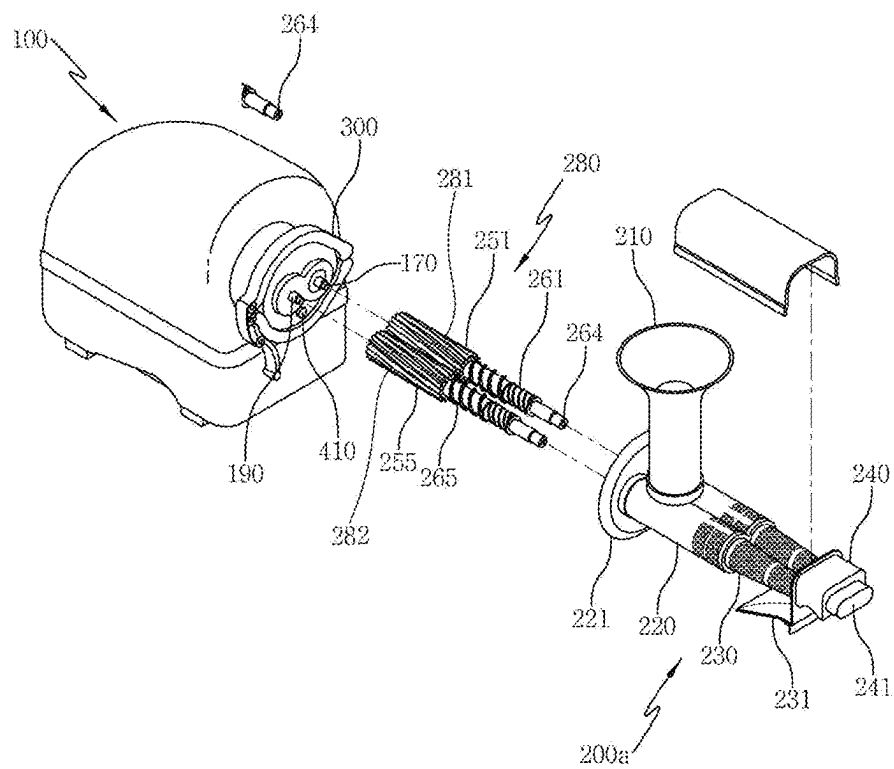


FIG. 4

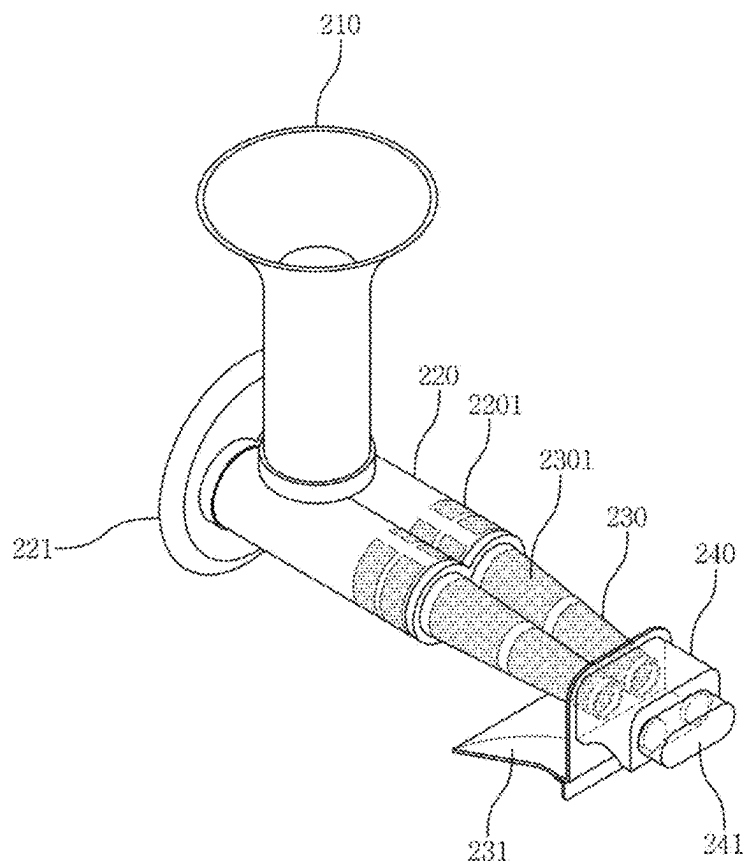


FIG. 5

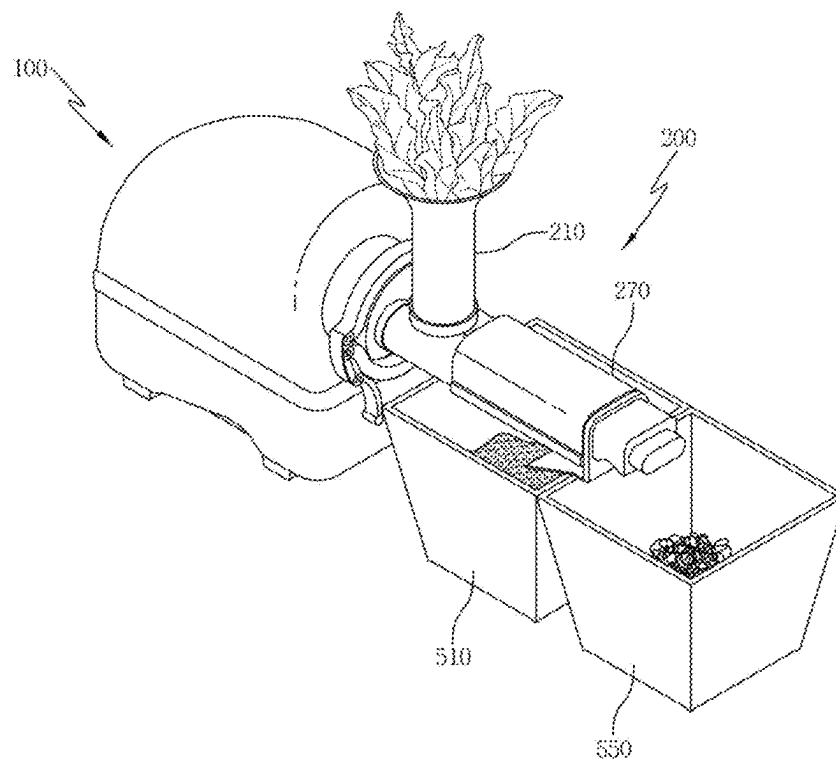


FIG. 6

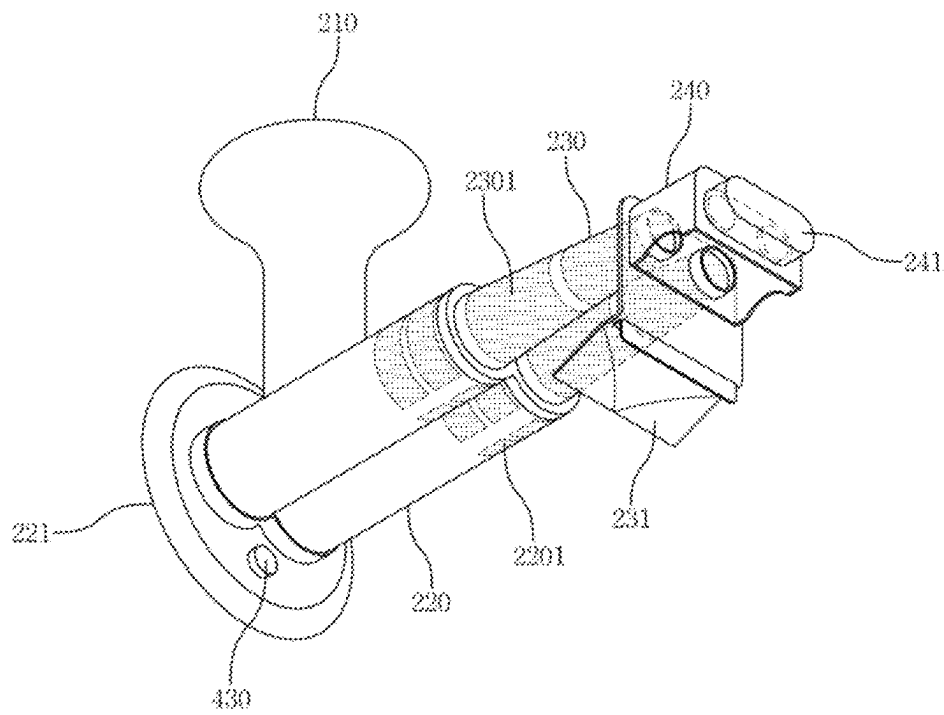


FIG. 7

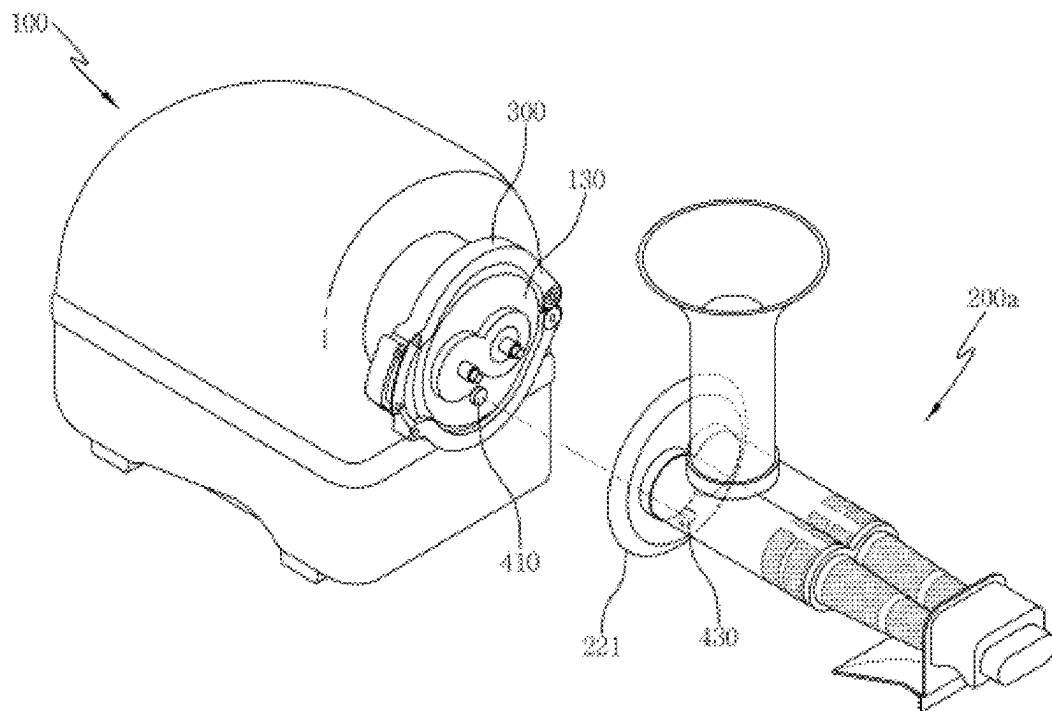


FIG. 8

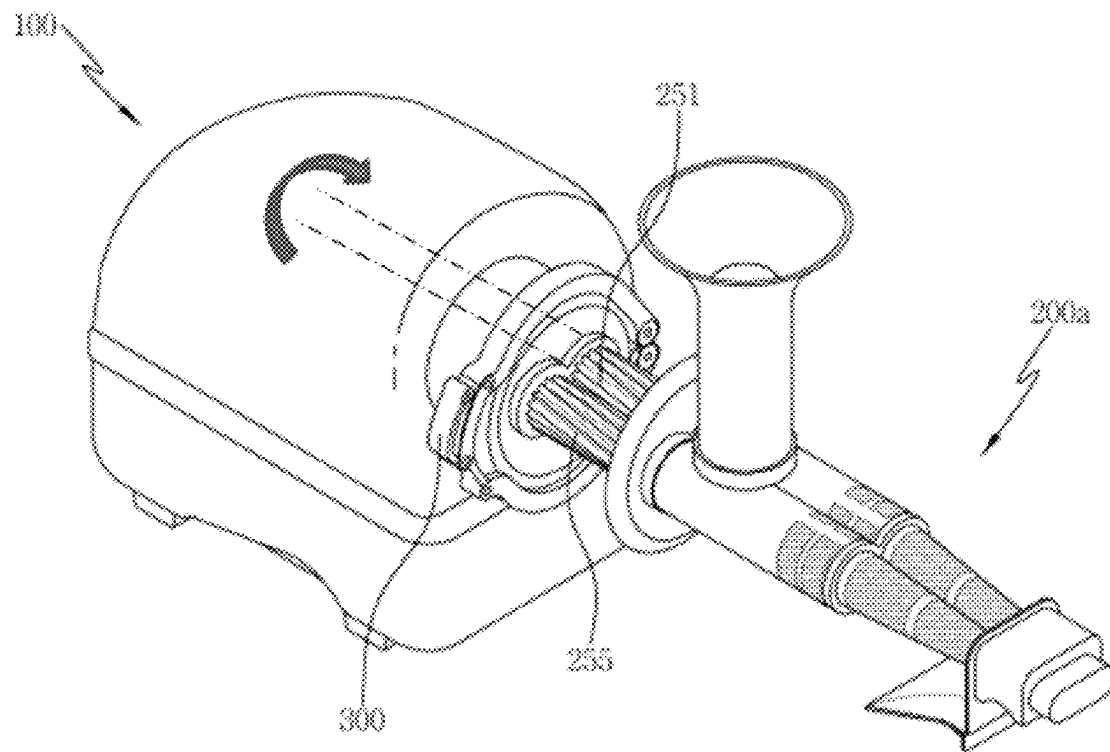
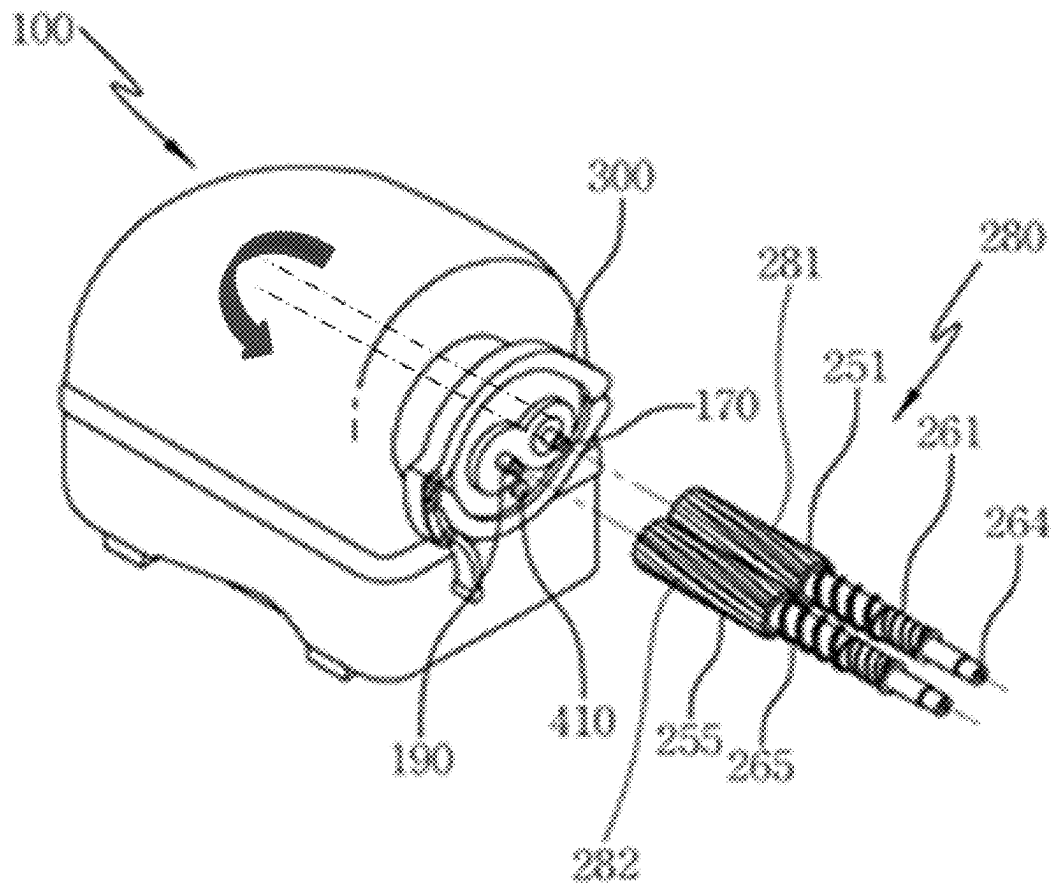


FIG. 9



1

JUICER**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to Korean Patent Application No. 10-2020-0027696, filed on Mar. 5, 2020, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present disclosure relates to a juicer, and more particularly, to a juicer with improved safety.

BACKGROUND ART

Generally, a juicer such as a juicer for green vegetable juice extracts juice by compressing various kinds of vegetables, fruits, or the like and mainly uses a centrifugal separation method and a pair gear method.

The centrifugal separation method is suitable for a juicer which extracts fruit juice, but has problems in that, even if a vegetable, which is much in fiber and less in moisture, is cut and rotated at a considerably high speed in order to extract juice, a juice extraction rate is not high and the residual heavy metal is discharged.

On the other hand, the pair gear method has a configuration of extracting cutting and juice by compressing the material or simultaneously cutting and compressing the material, and has advantages in that the juice extraction rate is several times higher than that of the centrifugal separation method and the residual heavy metal is not eluted together with the green vegetable juice, such that the juicer using the pair gear method occupies most of the current juicer market demand.

The juicer using the pair gear method includes a pair of grinding parts having a helical gear form and a pair of transfer parts having an Archimedes spiral form, and is configured by accommodating the grinding parts and the transfer parts within a housing in which a juice net is formed and then coupling the housing with a drive part through a fastening part.

The pair of grinding parts is driven by the drive part to transfer a material to the transfer parts while rotating with being engaged with each other, cutting, grinding, and juicing the put material, and the transfer parts compress the ground material while transferring the ground material, thereby extracting the juice.

However, the conventional juicer has a problem in that the grinding part and the drive part of the juicer are coupled to each other to drive the juicer uniformly, thereby causing a risk of safety accidents due to the unexpected operations of the grinding part and the drive part.

RELATED ART DOCUMENT

Patent Document

(Patent Document 1) Korean Patent No. 10-0334309

DISCLOSURE**Technical Problem**

The present disclosure is intended to solve the above problem, and an object of the present disclosure is to provide

2

a juicer with improved safety, in which a drive part and a roller part of a juicer are automatically attached and detached by the normal and reverse rotation of a motor, thereby preventing safety accidents.

Technical Solution

An embodiment of a juicer with improved safety according to the present disclosure may include:

- a motor housing;
 - a drive part including a motor which is provided inside the motor housing; and
 - a roller part which is detachably coupled to extend forward from the motor housing, and is provided therein to be rotatably driven by a driving force which is transferred from the drive part, and
- the drive part and the roller part may be mutually attached and detached by the normal and reverse rotation of the motor.

In an embodiment of the present disclosure, the drive part may be provided with a sensor operating part, and

- a roller part housing including the roller part may include a sensor part composed of a sensor which is operated by a change in a distance from the sensor operating part.

In an embodiment of the present disclosure, the roller part may be coupled to the drive part when the motor is rotated normally, and the roller part may be separated from the drive part when the motor is rotated reversely.

In an embodiment of the present disclosure, the motor may be automatically rotated normally if a fastening completion signal is received through a controller after the roller part is fastened to the drive part.

In an embodiment of the present disclosure, the motor may be automatically rotated reversely if a juicing completion signal is received through the controller after the driving of the roller part is terminated.

In an embodiment of the present disclosure, the roller part may include:

- a first gear roller which is connected to a rotary shaft of the driver part and gradually decreases in outer diameter in a discharge direction of sludge; and
- a second gear roller which is connected to a fixed shaft of the drive part and gradually decreases in outer diameter in the discharge direction of the sludge.

In an embodiment of the present disclosure, the juicer with the improved safety may further include sludge compressing force generator which generates a compressing force to sludge before a juicing material transferred between the outer surface of the roller part and the inner surface of the main housing is finally discharged.

In an embodiment of the present disclosure, the sludge compressing force generator may be provided at the end of the main housing, and is provided to be formed in a rim direction of a stepped part to give a resistance against a transferring force of the sludge, the stepped part being formed to be stepped between the sludge compressing force generator and an extension protruding from the front end of the roller part.

In an embodiment of the present disclosure, the motor may be a normal and reverse rotation motor, and the motor may be operated with a smaller output upon normal rotation than that upon reverse rotation.

Advantageous Effects

According to an embodiment of the juicer with improved safety according to the present disclosure, it is possible to

3

improve safety because a drive part and a roller part of a juicer are automatically attached and detached by the normal and reverse rotation of a motor, thereby preventing safety accidents.

DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective diagram illustrating an embodiment of a juicer with improved safety according to the present disclosure.

FIG. 2 is a vertical cross-section diagram of FIG. 1.

FIG. 3 is an exploded perspective diagram of FIG. 1.

FIG. 4 is a perspective diagram illustrating a housing among components illustrated in FIG. 1.

FIG. 5 is an external perspective diagram illustrating a juicing state using the juicer with improved safety according to the present disclosure.

FIG. 6 is a perspective diagram illustrating a state where a sensor operating means is provided in a housing 200a according to an embodiment of the present disclosure.

FIG. 7 is a perspective diagram illustrating for explaining a coupling process between a drive part 100 and the housing 200a according to an embodiment of the present disclosure.

FIGS. 8 and 9 are perspective diagrams illustrating a process of attaching and detaching the drive part 100 and the housing 200a of the juicer according to an embodiment of the present disclosure.

BEST MODE

Hereinafter, an embodiment of a juicer with improved juicing performance according to the present disclosure will be described in detail with reference to the accompanying drawings.

In adding reference numerals to the components of each drawing, it should be noted that the same components are denoted by the same reference numerals as possible even though they are illustrated in different drawings. Further, in describing embodiments of the present disclosure, if it is determined that the detailed description of the related publicly-known configuration or function disturbs the understanding of the embodiments of the present disclosure, the detailed description thereof will be omitted.

In describing the components of the embodiments of the present disclosure, terms such as first, second, A, B, (a), and (b) may be used. These terms are only to distinguish a component from another component, and the nature, order, sequence, and the like of the corresponding component are not limited by the terms. Further, unless otherwise defined, all terms used herein, including technical or scientific terms, have the same meaning as commonly understood by those skilled in the art to which the present disclosure pertains. Terms such as those defined in the commonly used dictionaries should be construed as having meanings consistent with the meanings in the context of the related art, and shall not be construed as ideal or excessively formal meanings unless expressly defined in this application.

FIG. 1 is a perspective diagram illustrating an embodiment of a juicer with improved juicing performance according to the present disclosure, FIG. 2 is a vertical cross-sectional diagram of FIG. 1, and FIG. 3 is an exploded perspective diagram of FIG. 1.

Hereinafter, in order to clearly understand the present disclosure, the description will be made by defining a horizontal direction of FIG. 2 as a “longitudinal direction”, a left direction of FIG. 2 as a rear side, a right direction of

4

FIG. 2 as a front side, a vertical direction of FIG. 2 as the top and the bottom, and a surface (paper) direction of FIG. 2 as a “width direction”.

As illustrated in FIGS. 1 to 3, an embodiment of the juicer with improved juicing performance according to the present disclosure includes a drive part 100 and a juicing part 200.

The drive part 100 serves to drive the juicing part 200, and to provide a driving force so as to drive the juicing part 200, into which a material such as a vegetable or a fruit (hereinafter, referred to as “juicing material”) is put, to grind and press the juicing material, thereby extracting the juice. The juicing part 200 includes a substantial juicing space where the juicing material is juiced while being put and the residual material is discharged to the outside in the form of sludge. The drive part 100 and the juicing part 200 may be firmly, detachably coupled by a fastening part 300 to be described later.

As illustrated in FIGS. 1 to 3, the drive part 100 may be configured to include a motor housing 110 which has a motor 150 provided therein, and has a hollow first connecting part 130 which protrudes forward and whose end increases and then decreases in diameter forward in the longitudinal direction, a rotary shaft 170 which rotates with one end connected to the motor 150 and the other end protruding in the longitudinal direction through the first connecting part 130, and a fixed shaft 190 which is provided to protrude from the first connecting part 130 in parallel with the rotary shaft 170. The rotary shaft 170 may be connected to the motor 150 through a speed reducer not illustrated.

The front end of the first connecting part 130 having a plate shape has an occluded structure with the protruding central portion. The protruding central portion has a form in which parts of two circular arcs overlap each other in the width direction, the fixed shaft 190 may be formed to protrude from the center of the circular arc at one side in the width direction, and the rotary shaft 170 may protrude through the center of the circular arc at the other side in the width direction.

As illustrated in FIGS. 1 to 3, the juicing part 200 may be composed of the roller part 280 which is rotatably connected to the drive part 100, and a hollow main housing 200a which accommodates the roller part 280 and is coupled to the motor housing 110 of the drive part 100.

The roller part 280 may include a grinding roller part 250, and a juicing roller part 260 which is connected to the front of the grinding roller part 250 and extends to decrease in diameter forward. The roller part 280 serves to grind the juicing material between the outer circumferential surface of the roller part 280 and the inner surface of the main housing 200a while being accommodated in the main housing 200a to rotate, and to press and juice the ground juicing material to transfer the material.

Here, the roller part 280 may be attached and detached to and from the drive part 100 automatically or manually by the normal and reverse rotation of the motor 150, which will be described in more detail with reference to the drawings below.

As illustrated in FIGS. 2 and 3, the grinding roller part 250 may have a grinding blade formed on the outer circumferential surface thereof, and include a first grinding roller 251 which is axially coupled to rotate integrally with the rotary shaft 170, and a second grinding roller 255 which is rotatably coupled to the fixed shaft 190.

The rotary shaft 170 may have a spline shape on the outer circumferential surface thereof or a polygonal cross section, and may be formed with a concave groove having a polygo-

5

nal cross section so that the rotary shaft **170** may be inserted into the center of the rear end of the first grinding roller **251** or engaged with a spline.

The fixed shaft **190** may be formed to have a circular cross section shape, and formed with a concave groove having a circular cross section so that the fixed shaft **190** is inserted into and rotatably supported by the second grinding roller **255**, and the second grinding roller **255** may idle by being rotatably supported by the fixed shaft **190**. The first grinding roller **251** and the second grinding roller **255** may have a protrusion which protrudes forward or may be formed with a concave groove in the front surface thereof.

The grinding blades formed on the outer circumferential surfaces of the first grinding roller **251** and the second grinding roller **255** are provided to have a gear shape (preferably, a helical gear shape) to be engaged with each other, such that when the first grinding roller **251** coupled to the rotary shaft **170** rotates integrally with the rotary shaft **170**, the second grinding roller **255** coupled to the fixed shaft **190** so as to idle by the engaged grinding blade may also rotate together.

Further, the juicing material is transferred forward while being ground and juiced by the grinding blades engaged with each other to rotate. When viewed from the front to the rear in FIG. 3, the first grinding roller **251** inserted into the rotary shaft **170** rotates counterclockwise, and the second grinding roller **255** rotates clockwise by the first grinding roller **251**, such that the juicing material put from the top toward the first grinding roller **251** and the second grinding roller **255** is transferred while being ground by the grinding blades engaged with each other to rotate. At this time, the grinding blades formed on the outer circumferential surfaces of the first grinding roller **251** and the second grinding roller **255** and engaged with each other have a helical gear form and form a substantially 'V' shape forward when viewed from the top, such that the put juicing material is transferred forward while being ground by the grinding blades. The put juicing material is preferably ground and quickly transferred forward by forming a lead angle of the grinding blade to be smaller than 30 degrees.

As illustrated in FIGS. 2 and 3, the juicing roller part **260** may include a first juicing roller **261** and a second juicing roller **265** which are formed with protrusions spirally protruding from the outer circumferential surfaces thereof.

The first juicing roller **261** may be formed integrally at the front end of the first grinding roller **251**, and the second juicing roller **265** may be formed integrally at the front end of the second grinding roller **255**. As described above, in an embodiment of the juicer with improved juicing performance according to the present disclosure, the roller part of the juicing part **200** configured to juice the juicing material substantially may be provided in a pair gear method having two gears: one gear composed of the first grinding roller **251** and the first juicing roller **261** and one gear composed of the second grinding roller **255** and the second juicing roller **265**.

Hereinafter, a configuration in which the first juicing roller **261** is coupled to the first grinding roller **251** to rotate integrally, and the second juicing roller **265** is coupled to the second grinding roller **255** to rotate integrally will be described.

Further, a configuration of the grinding roller part **250** and the juicing roller part **260** connected to the rotary shaft **170** of the drive part **100**, among the components of the roller part **280**, will be referred to as a first gear roller **281**, and a configuration of the grinding roller part **250** and the juicing roller part **260** connected to the fixed shaft **190** of the drive

6

part **100**, among the components of the roller part **280**, will be referred to as a second gear roller **282**.

The first grinding roller **251** and the second grinding roller **255** may have protrusions which protrude forward, or may be concavely formed with grooves so that the first juicing roller **261** and the second juicing roller **265** are coupled thereto, respectively. The first juicing roller **261** and the second juicing roller **265** may be formed at the rear of the first grinding roller **251** and the second grinding roller **255** with grooves into which the protrusions provided on the first grinding roller **251** and the second grinding roller **255** are inserted, or may have the protrusions which are inserted into the concave grooves formed on the fronts of the first grinding roller **251** and the second grinding roller **255** and may be coupled to the first grinding roller **251** and the second grinding roller **255**, respectively. At this time, the first grinding roller **251** and the first juicing roller **261** are coupled to each other in a forcibly fitting manner to rotate integrally, and the second grinding roller **255** and the second juicing roller **265** are coupled to each other in a forcibly fitting manner to rotate integrally. Of course, as described above, the first juicing roller **261** may also be formed integrally with the first grinding roller **251**, and the second juicing roller **265** may also be formed integrally with the second grinding roller **255**.

As illustrated in FIGS. 2 and 3, the first juicing roller **261** and the second juicing roller **265** of the juicing roller part **260** are formed to have a tapered shape in which outer circumferential surfaces **266** decrease in diameter (outer diameter) toward the front thereof. A spiral protrusion may be formed on each of the outer circumferential surfaces of the first juicing roller **261** and the second juicing roller **265**.

Each of the front ends of the first juicing roller **261** of the first gear roller **281** and the second juicing roller **265** of the second gear roller **282** is not formed with the aforementioned spiral protrusion and may be further formed with an extension **264** including a stepped part **284** which is formed with a stepped surface with a relatively decreased diameter. The end of the extension **264** is rotatably supported by a support through hole (with no reference numeral) formed in an accommodating part **241** of a discharge housing **240** to be described later.

FIG. 4 is a perspective diagram illustrating the main housing among the components illustrated in FIG. 1, and FIG. 5 is an external perspective diagram illustrating a juicing state using the juicer with improved juicing performance according to the present disclosure.

As illustrated in FIGS. 2 to 4, the main housing **200a** may be composed of a hollow grinding housing part **220**, and a hollow juicing housing part **230** which is branched from the grinding housing part **220** to extend to be tapered to decrease in diameter forward in the longitudinal direction.

The cross section of the grinding housing part **220** has a form in which parts of two circular arcs overlap each other so that the grinding blades of the first grinding roller **251** and the second grinding roller **255** may be inserted while being engaged with each other. A hollow hopper part **210**, which is formed to extend upward to put the juicing material into the grinding housing part **220**, may be provided at the top of the grinding housing part **220**, and the grinding housing part **220** of the portion where the hopper part **210** is provided is formed with a through hole so that the hopper part **210** and the grinding housing part **220** may communicate with each other.

As illustrated in FIG. 4, a first juicing hole **2201**, which is a plurality of fine through holes, may be formed in the grinding housing part **220** in front of the hopper part **210**.

When the juicing material is put into the hopper part **210**, the juicing material is transferred forward while being ground by the grinding blades of the first grinding roller **251** and the second grinding roller **255** which are engaged with each other to rotate, and a part of the juice flowing out from the juicing material while the juicing material is ground may be first discharged through the first juicing hole **2201**.

The juicing housing part **230** may be formed with a second juicing hole **2301**, which is a plurality of fine through holes, and a plate-shaped juicing solution discharge guider **231** is positioned to be inclined downward under the juicing housing part **230**. The juicing solution discharge guider **231** is formed integrally in the discharge housing **240** to be described later, and serves to guide the juicing solution of the juicing material juiced by the juicing part **200** to be discharged downward.

As described above, the juicing material transferred forward by the first grinding roller **251** and the second grinding roller **255** is primarily discharged as a juicing solution through the first juicing hole **2201**, and the juicing material which is not completely juiced is secondarily discharged as a juicing solution through the second juicing hole **2301** while being transferred toward the first juicing roller **261** and the second juicing roller **265** to be juiced. Here, before processing the secondary discharge, the compressing force which is transferred between the outer surface of the second juicing roller and the inner surface of the main housing **220a** (the second juicing hole **2301**) is finally discharged, through the sludge compressing force generator. The sludge compressing force generator has the end of the main housing **220a**, but may be formed in the rim direction of the stepped part to give the resistance against the transferring force.

As illustrated in FIGS. 3 and 4, the discharge housing **240** may be provided at the longitudinal end of the juicing housing part **230**. One end (rear end) of the discharge housing **240** may be coupled to the juicing housing part **230**, and the other end (front end) side thereof may be provided with the accommodating part **241**. The accommodating part **241** is provided at a position which is spaced at a predetermined distance in the longitudinal direction apart from one end (rear end) of the discharge housing **240** coupled to the juicing housing part **230**.

The extension **264**, which is provided at any one end of the first juicing roller **261** and the second juicing roller **265** accommodated in the juicing housing part **230**, may extend to protrude forward from the end of the juicing housing part **230** and may be inserted into and supported by the accommodating part **241** through a part of the discharge housing **240** coupled to the juicing housing part **230**.

More specifically, the extensions **264**, which are provided at the respective ends of the first juicing roller **261** of the first gear roller **281** and the second juicing roller **265** of the second gear roller **282**, may be configured so that a first extension **264** of the first juicing roller **261** and a second extension **264** of the second juicing roller **265** are rotatably inserted into and supported by two support through holes which are provided in the accommodating part **241** in the width direction, respectively.

The discharge housing **240** may have an opening formed downward between one end coupled to the juicing housing part **230** and the accommodating part **241**, so that the sludge, which is the residue of the juicing material after the juicing material is juiced, is discharged. Hereinafter, the description by defining, as 'discharge direction', the direction in which the sludge is discharged in a longitudinal direction of the main housing **200a**.

As illustrated in FIGS. 4 and 5, the juicing solution discharge guider **231** is formed integrally with the discharge housing **240**, as described above, and is formed to be inclined downward from the rear coupled to the juicing housing part **230** and positioned under the juicing housing part **230**. Accordingly, a part of the juice which is juiced by the juicing roller part **260** and discharged through the second juicing hole **2301** falls to the juicing solution discharge guider **231** to flow downward.

FIG. 6 is a perspective diagram illustrating a state where a sensor operating means is provided in the housing **200a** according to an embodiment of the present disclosure, FIG. 7 is a perspective diagram illustrating for explaining a coupling process between the drive part **100** and the housing **200a** according to an embodiment of the present disclosure, and FIGS. 8 and 9 are diagrams illustrating a state where the drive part and the roller part of the juicer according to an embodiment of the present disclosure are detached.

A sensor part **400** is provided in the drive part **100** and the juicing part **200** according to the present disclosure, and the sensor part **400** may include a sensor **410** which is provided at one side of the first connecting part **130** of the motor housing **110**, and a sensor operating part **430** which is provided in a second connecting part **221** of the housing **200a**. As the sensor part **400**, for example, various types of proximity sensors may be used, which operate depending upon whether an object approaches another object, such as a magnetic proximity sensor, an optical proximity sensor, or an ultrasonic proximity sensor.

Here, describing the use of the magnetic proximity sensor, the sensor **410** in the present disclosure may be a reed switch operated by a magnet, and the sensor operating part **430** may be a permanent magnet.

The sensor **410** may become an 'ON' state when approaching the sensor operating part **430**, and may become an 'OFF' state when moving away from the sensor operating part **430**. The sensor **410** may be provided on a path of connecting a motor with a power source. When the juicing part **200** is coupled to the drive part **100** through the fastening part **300**, the sensor **410** and the sensor operating part **430** are provided to face in proximity to each other, such that the sensor **410** becomes an 'ON' state, and when a switch not illustrated is turned on, power is supplied to the motor **150** to drive the motor **150**, and the first grinding roller **251**, the second grinding roller **255**, the first juicing roller **261**, and the second juicing roller **265** may rotate within the housing **200a** through the rotary shaft **170**.

According to the present disclosure, the motor **150** may be automatically turned on/off according to the operating states of the sensor part **400** and the sensor operating part **430** of the motor housing **110**. That is, when the sensor **410** approaches the sensor operating part **430** and becomes an 'ON' state, the motor **150** automatically becomes the 'ON' state to operate, and when the sensor **410** moves away from the sensor operating part **430** and becomes an 'OFF' state, the motor **150** becomes an 'OFF' state to automatically stop.

Furthermore, the motor **150** may be operated/stopped by the sensor part **400** and the sensor operating part **430**, and the drive part **100** and the roller part **280** are mutually attachable to and detachable from each other by the normal and reverse rotation of the motor **150**. That is, as illustrated in FIGS. 8 and 9, when the roller part **280** is coupled to the drive part **100**, the motor **150** may be rotated normally such that the roller part **280** and the drive part **100** are tightly fastened to each other, and when the roller part **280** prepares to be separated from the drive part **100**, the motor **150** may be

rotated reversely such that the roller part **280** and the drive part **100** are released from being coupled to each other.

According to the present disclosure, the motor **150** may be operated and stopped by the sensor part **400** and the sensor operating part **430** and the operations of the motor **150** may be controlled through the controller (not illustrated). That is, a separation distance between the roller part **280** and the drive part **100** is sensed by the sensor part **400** and the sensor operating part **430**, such that the operation/stop of the motor **150** may be prepared and in conjunction therewith, the motor **150** may be normally or reversely rotated automatically or manually through the controller and the roller part **280** may be attached and detached from the drive part **100**.

Specifically, as described above, when the sensor **410** approaches the sensor operating part **430** and becomes the 'ON' state, the motor **150** automatically becomes the 'ON' state according to a signal of the controller to operate. At this time, when the user couples the roller part **280** with the drive part **100**, the motor **150** is rotated normally by an operating signal of the controller, such that the roller part **180** may be tightly coupled to the drive part **100**. Conversely, when the sensor **410** moves away from the sensor operating part **430** and becomes the "OFF" state, the motor **150** may become the "OFF" state to automatically stop. At this time, when the user wants to separate the roller part **280** from the drive part **100**, the motor **150** is reversely rotated by the operating signal of the controller, such that the roller part **180** may be removed from the drive part **100** more easily.

Accordingly, by controlling, by the controller, the motor **150** to be normally and reversely rotated automatically or manually, the controller controlling the operation of the motor **150** according to the proximity distance between the sensor **410** and the sensor operating part **430** and the attachment and detachment of the roller part **280** to and from the drive part **100**, it is possible to fundamentally prevent safety accidents capable of occurring when the user detaches the roller part **280** from the drive part **100**.

As described above, the embodiment of the juicer with improved juicing performance according to the present disclosure has been described in detail with reference to the accompanying drawings. However, the embodiment of the present disclosure is not necessarily limited to the aforementioned embodiment, and it is natural that various modifications and equivalents thereof may be made by those skilled in the art to which the present disclosure pertains. Accordingly, the true scope of the present disclosure will be defined by the claims to be described later.

The invention claimed is:

1. A juicer with improved safety comprising:

a motor housing;

a drive part comprising a motor which is provided inside the motor housing, wherein the drive part is provided with a sensor operating part;

a roller part which is detachably coupled to the motor housing and extending forward therefrom, and to be rotatably driven by a driving force which is transferred from the drive part; and

a sensor part composed of a sensor which is operated by a change in a distance from the sensor operating part, wherein a roller part housing comprising the roller part comprises the sensor part;

wherein the motor is configured to be controlled according to a signal generated based on the change in the distance between the sensor and the sensor operating part and an attachment and detachment of the roller part to the drive part,

wherein the drive part and the roller part are mutually attached and detached by normal and reverse rotation of the motor,

wherein, when the sensor operating part approaches the sensor, the motor is in an "ON" state, and when the roller part is coupled to the drive part when it is in the "ON" state, the motor rotates forward according to an operation signal and the roller part is coupled to the drive part,

and when the sensor operating part moves away from the sensor and becomes the "OFF" state, the motor automatically becomes the "OFF" state according to the operating signal, and when in the "OFF" state, the roller part rotates in reverse according to the operation signal and is separated from the drive part,

wherein the motor is automatically rotated normally if a fastening completion signal is received after the roller part is fastened to the drive part, and the motor is automatically rotated reversely if a juicing completion signal is received after the driving of the roller part is terminated such that the roller part is separated from the drive part, and

wherein the motor is capable of rotating both in normal and reverse directions, and the motor operates with a smaller output during the normal rotation than during the reverse rotation.

2. The juicer of claim 1,

wherein the roller part comprises:

a first gear roller which is connected to a rotary shaft of the driver part and gradually decreases in outer diameter in a discharge direction of sludge; and

a second gear roller which is connected to a fixed shaft of the drive part and gradually decreases in outer diameter in the discharge direction of the sludge.

3. The juicer of claim 1, further comprising a sludge compressing force generator which generates a compressing force to sludge before a juicing material which is transferred between the outer surface of the roller part and the inner surface of the main housing is finally discharged.

4. The juicer of claim 3,

wherein the sludge compressing force generator is provided at the end of the main housing, and is provided to be formed in a rim direction of a stepped part to give a resistance against a transferring force of the sludge, the stepped part being formed to be stepped between the sludge compressing force generator and an extension protruding from the front end of the roller part.

* * * * *